

Based on the provided sources, load balancing is the process of distributing workloads across multiple virtual machines (VMs) or physical servers to optimize resource utilization and prevent system failures 1, 2. The types of load balancing are primarily categorized by their logic (Static vs. Dynamic) and their architectural scope.

1. Logic-Based Types

Static Load Balancing

Static load balancing uses predefined rules to distribute tasks without considering the real-time state of the system 3.

- **Characteristics:** Simple to implement and effective for uniform workloads, but it does not account for differences in VM capacity or current usage 3.
- **Algorithms:**
- **Round Robin:** Assigns tasks in a cyclic manner 3.
- **Weighted Round Robin:** Assigns weights based on processing power, making it better for heterogeneous environments 3.
- **Randomized Algorithm:** Assigns tasks randomly; suitable for high volumes of requests 3.

Dynamic Load Balancing

Dynamic load balancing adjusts task assignments in real-time based on current system conditions 4.

- **Characteristics:** It is ideal for variable workloads and helps ensure VMs are not assigned more tasks than they can handle 4, 5. However, it adds computational overhead due to the need for continuous monitoring 6.
- **Algorithms:**
- **Least Connection:** Assigns tasks to the VM with the fewest active connections; ideal for web servers 4, 7.
- **Throttled:** Ensures a VM does not exceed its capacity through constant load monitoring 4.
- **Min-Min and Max-Min:** Prioritize smaller tasks and larger tasks, respectively 4, 8.
- **Intelligent Algorithms:** Includes **Ant Colony Optimization (ACO)**, **Artificial Neural Networks (ANN)**, and **Honeybee Foraging**, which use probabilistic or predictive methods 8.

2. Architectural Types

While the sources do not explicitly use the terms "centralized" and "distributed" as formal headers for types, they describe these architectures through the following classifications:

Centralized Load Balancing (Implied)

In this architecture, a dedicated entity acts as a single point of entry to manage and distribute traffic.

- **Hardware-Based:** Uses dedicated devices like F5 BIG-IP 9.
- **Software-Based:** Uses applications like NGINX or AWS Elastic Load Balancer (ELB) placed in front of VMs to monitor and distribute requests 5, 9.

Distributed Load Balancing (Global/DNS)

This architecture spreads the load-balancing function across multiple locations or systems.

- **Global Server Load Balancing (GSLB):** Distributes traffic across multiple data centers in different geographic regions to reduce latency (e.g., Google or Netflix) 10.
- **DNS Load Balancing:** Distributes traffic by resolving domain names to different IP addresses associated with servers in various locations 10.

3. Comparison Table

Feature,	Static Load Balancing,	Dynamic Load Balancing
Decision Base,	Predefined rules/logic 3.,	Real-time system state 4.
Complexity,	Low (simple to implement) 3.,	High (requires continuous monitoring) 6.
Resource Awareness,	None; ignores current load 3.,	"High; monitors VM capacity 4, 5."
Best For,"	"Uniform, predictable workloads 3.",	"Variable, heterogeneous workloads 4, 7."
Overhead,	Minimal 3.,	Noticeable computational overhead 6.

4. Exam-Ready Explanation

Load balancing is vital in virtualization because VMs share physical resources (CPU, RAM); efficient distribution ensures no single resource is overwhelmed 2.

Key Categorization:

1. **Logic:** Choose **Static** (e.g., Round Robin) for simplicity when all servers are identical. Choose **Dynamic** (e.g., Least Connection) for enterprise environments like **Amazon** to handle traffic surges and ensure high availability 5, 7.
2. **Infrastructure:** Use **Hardware-based** for massive data centers or **Software-based** (like ELB) for cloud scalability 9. For global applications, **GSLB** is used to route users to the nearest regional data center 10.

Real-Life Impact: Before implementation, e-commerce sites may face 8–10 second load times and server crashes. After implementing load balancing with algorithms like Least Connection, response times can drop to 2–3 seconds with 24/7 uptime and a significant increase in completed transactions 11.